

- Document Control
- System Name: Application
- Version: 1.0
- Author: Data Engineering Team
- Date: 2024-06-12
- Status: Draft

- Executive Summary
 - The Application facilitates an automated ETL pipeline involving source CSV data ingestion, transformation, cleaning, aggregation, and SCD Type 2 history management, implemented using Delta Live Tables.

- Business Objectives
 - Automate data ingestion from CSV files to Delta tables
 - Ensure data quality through null and duplicate removal
 - Aggregate customer spend and maintain accurate, auditable transaction history
 - Enable change data capture via SCD Type 2 implementation

- In-Scope
 - Source CSV data reading and delta table loading
 - Schema enforcement for customer and order tables
 - Data transformation and cleaning logic
 - Creation of ordersummary and customeraggregatespend tables
 - Aggregation logic on ordersummary data
 - SCD Type 2 implementation for ordersummary
 - Process deployment using Delta Live Tables

- Out-of-Scope
 - Any manual data manipulation
 - External system integrations beyond defined delta tables
 - Real-time or streaming ingestion
 - Data visualization or reporting dashboards

- Stakeholders
 - Data Engineering Team
 - Business Analysts
 - Compliance Team

- Business Requirements

BR-ETL-001 — Load Source Customer Data

- Description: Read customer CSV data and load to delta table /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Business Value: Automated, structured customer data ingestion
- Priority: High

BR-ETL-002 — Load Source Order Data

- Description: Read order CSV data and load to delta table /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Business Value: Automated, structured order data ingestion
- Priority: High

BR-ETL-003 — Customer Table Schema Enforcement

- Description: Enforce schema with fields CustId, Name, EmailId, Region for customer table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-004 — Order Table Schema Enforcement

- Description: Enforce schema with fields OrderId, ItemName, PricePerUnit, Qty, Date, CustId for order table
- Business Value: Consistency and data quality assurance
- Priority: High

BR-ETL-005 — Calculate TotalAmount Field

- Description: Add column TotalAmount to order table ($\text{PricePerUnit} * \text{Qty}$)
- Business Value: Simplifies downstream aggregation and analytics
- Priority: Medium

BR-ETL-006 — Data Quality: Null and Duplicate Removal

- Description: Remove nulls and duplicates from customer and order tables
- Business Value: Improves analytical accuracy and reliability
- Priority: High

BR-ETL-007 — Create Ordersummary Table

- Description: Create ordersummary table in catalog=gen_ai_poc_databrickscoe, schema=sdlc_wizard if not exists
- Business Value: Centralized storage for enriched order data
- Priority: High

BR-ETL-008 — SCD Type 2 on Ordersummary

- Description: Apply SCD Type 2 methodology to ordersummary based on updates in customer table
- Business Value: Ensures historical accuracy and traceability of data changes
- Priority: High

BR-ETL-009 — Create CustomerAggregateSpend Table

- Description: Create customeraggregatespend table with columns Name, TotalAmount, Date if not exists
- Business Value: Enables spend analysis and reporting
- Priority: Medium

BR-ETL-010 — Aggregate Customer Spend

- Description: Aggregate TotalAmount grouped by Name and Date from ordersummary and load into customeraggregatespend
- Business Value: Facilitates spend tracking for business insights
- Priority: Medium

BR-ETL-011 — Delta Live Tables Implementation

- Description: Implement the entire processing steps using Delta Live Tables
- Business Value: Simplifies workflow orchestration and monitoring
- Priority: High
- Assumptions & Constraints
 - Source data is available and accessible via defined CSV file paths
 - Only the specified schemas will be enforced
 - Delta Lake infrastructure is operational
- Success Metrics
 - Successful, automated runs from CSV to target tables
 - Zero null/duplicate records post-processing
 - Accurate aggregation results in customeraggregatespend
 - SCD Type 2 audit trails present
- Risks & Mitigations
 - Source data format changes — Mitigation: Schema enforcement and validation
 - Data quality issues from source — Mitigation: Robust null and duplicate handling logic
 - Infrastructure/permission errors — Mitigation: Pre-execution checks and monitoring

- Approval
 - Business Owner: Data Analytics Director
 - Data Engineering Lead: Lead Data Engineer
 - Compliance Review: Compliance Officer
 - Date: 2024-06-12
-
- Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Source customer input data
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Source order input data

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Processed	ordersummary	OrderId, ItemName, PricePerUnit, Qty, Date, CustId, TotalAmount, scd_fields	Enriched
Aggregatespend	Processed	customeraggregatespend	Name, TotalAmount, Date	Aggregated

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Calculate TotalAmount	orderdata	ordersummary	TotalAmount = PricePerUnit * Qty
2	Remove Nulls	customerdata, orderdata	cleaned tables	Filter out records with nulls
3	Remove Duplicates	customerdata, orderdata	deduped tables	Drop duplicate records
4	SCD Type 2	customerdata, ordersummary	ordersummary	Maintain change history on customer updates
5	Aggregate Spend	ordersummary	customeraggregatespend	Sum TotalAmount by Name, Date

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-ETL-003	Customer Data	Enforce schema for customer table	Requirements
BR-ETL-004	Order Data	Enforce schema for order table	Requirements
BR-ETL-006	Data Quality	Remove nulls and duplicates from tables	Requirements
BR-ETL-008	SCD Handling	Apply SCD Type 2 for order summary	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Load customer and order CSVs into delta tables	Source data accessible

2	Clean and deduplicate input tables	Initial load complete
3	Perform transformations including TotalAmount calculation	Data cleaned
4	Aggregate and load to target tables	Transformations complete
5	Maintain historical changes in ordersummary	Customer updates